Online changepoint detection using forecasts

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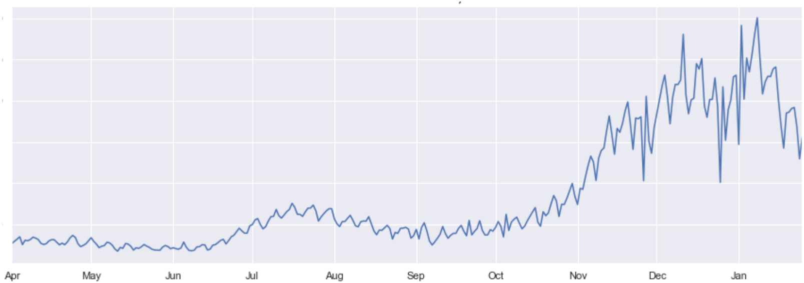
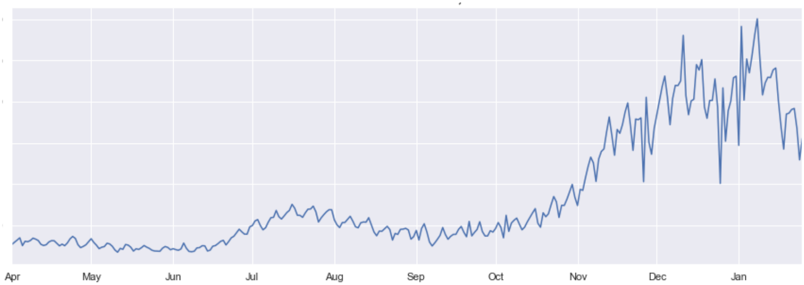
Motivation



Motivation



- Multiple seasonal frequencies
- Regressors for holiday periods
- Explanatory series
- Lagged dependencies

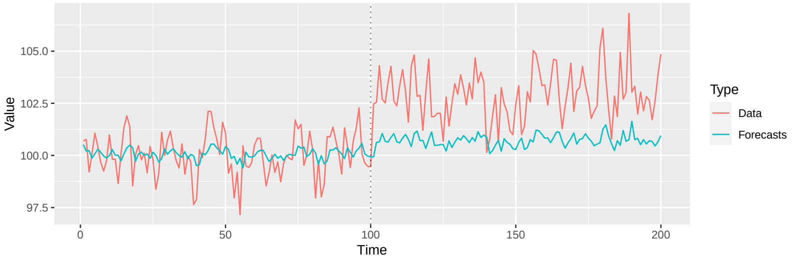
Problems

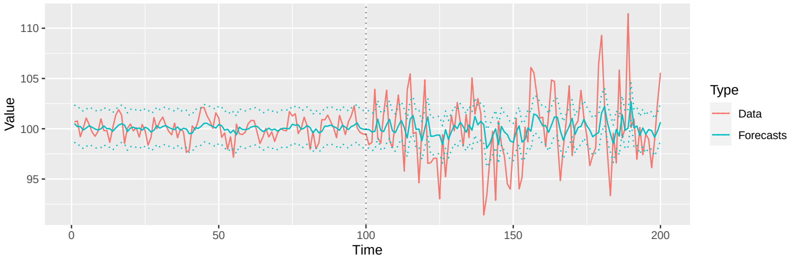
Automatically identifying changes in data requiring complex modelling is problematic:

- Most methods need to be able to fit the model after the change ...
- ... leading to delays in online detection
- ... poor model estimation in short segments
- ... identifiability issues with changepoints close together
- ... inability to fit long seasonal frequencies (e.g. yearly)

- Those that don't are often restricted to mean/linear trend series only.

Using Forecasts





Causes of Poor Forecasts

A change in the underlying process (data) could cause our forecasting model to behave poorly!

Intuitively:

- Mean change in raw data $\rightarrow$ forecasts become biased.
- Variance change in raw data $\rightarrow$ prediction intervals become poor.
- Other changes in raw data $\rightarrow$ potential combination of biased and inaccurate prediction intervals.

Our Aim

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Aim: Create a framework for detecting changes in complex models which doesn't require fitting the model to the post-change data.

Intuitively, changes in the raw data = forecasts become poor.

Solution: Use sequential forecast errors.

Bonus: This also provides a framework for identifying changes in forecast performance.

Detecting Changes

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Detector:

- Measures the difference between mean in training period and mean in monitoring period.
- CUSUM Detector: $Q(m,k) = \sum_{t=m+1}^{k} y_t - \frac{k}{m}\sum_{t=1}^{m} y_t.$
- Page's CUSUM: $D(m,k) = \max_{0\leq t\leq k} |Q(m,k) - Q(m,t)|$

Stopping Rule:

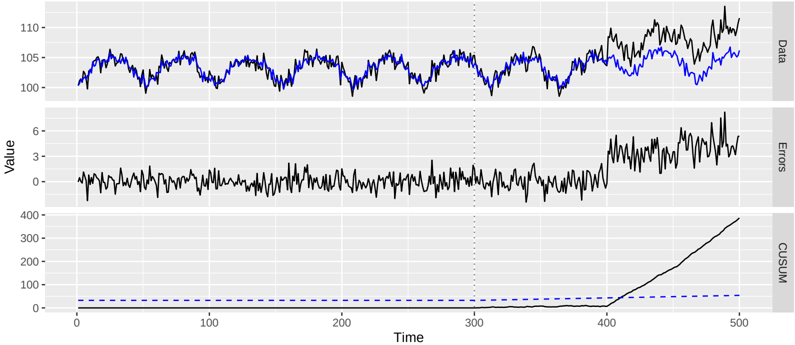
- Defines a rule for when to flag a change.
- Typically when the detector exceeds some threshold.
- Threshold controls false positive rate.
- $\tau = \min\{k \geq 1 : D(m,k) \geq c\hat{\sigma}_m g(m,k)\}.$

Solution

What about the forecast errors?

$$e_t = Y_t - \hat{y}_t(1)\ , \qquad Q(m,k) = \sum_{t=m+1}^{k} e_t - \frac{k}{m}\sum_{t=1}^{m} e_t$$

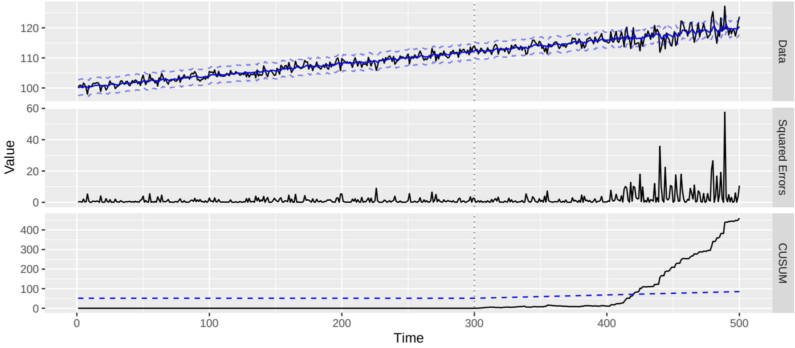
- Forecasting model accounts for data complexities.



Variance Changes

Variance change in errors = Mean change in squared errors.

$$Q(m,k) = \sum_{t=m+1}^{k} (e_t - \mu_m)^2 - \frac{k}{m}\sum_{t=1}^{m}(e_t - \mu_m)^2 \ .$$



Theory & Simulations

Theory: Under certain assumptions,

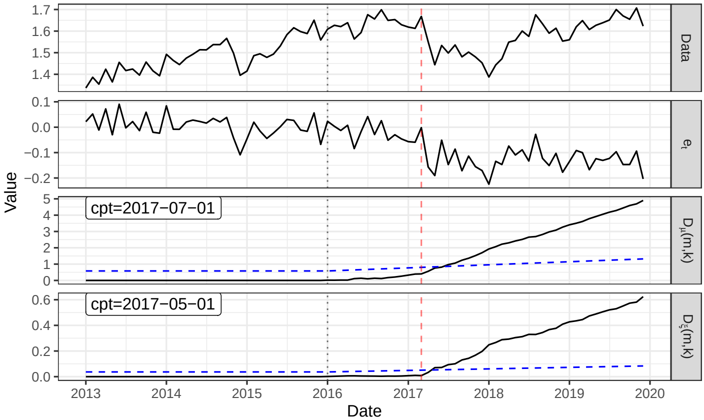
- Mean change in raw data $\rightarrow$ mean change in forecast errors.
- Mean and/or variance change in raw data $\rightarrow$ mean change in squared forecast errors.

Simulations:

- Using forecast errors performs better than raw CUSUM.
- Can also be used to detect changes in,
  - Trend.
  - Dependence structure.
  - Error structure.

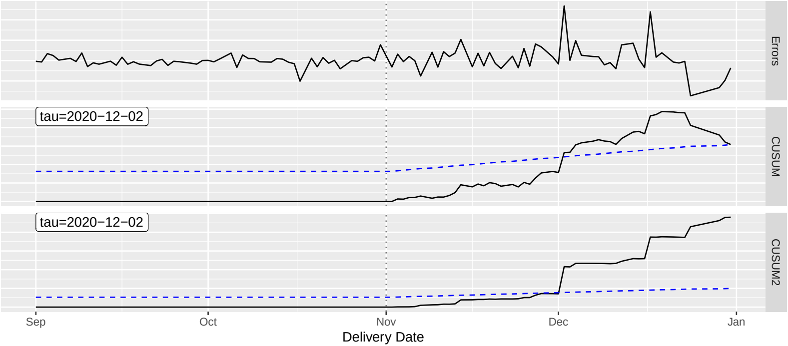
A&E Admissions

- A&E Admissions with Gallstone related pathology
- Predicting admissions helps to manage workload
- Elective surgeries can be scheduled for quieter periods



TTP Project

- Forecast the no. of parcels being delivered from each delivery office.
- In each year and delivery office the Xmas rush starts at different times.
- We can use this framework to detect when the Xmas rush begins.



TTP Project

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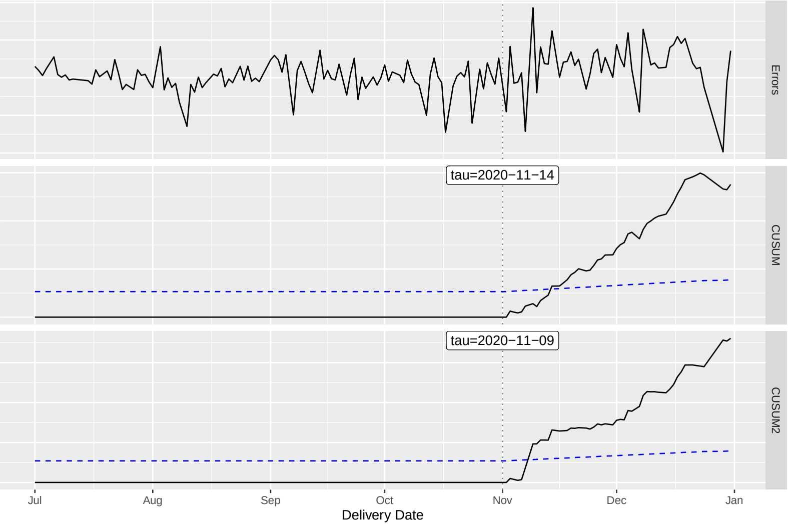

Conclusion

Summary:

- Created a framework for detecting when/if forecasts become poor.
- Shown using forecast errors is better than the raw data for detecting changes.

Future work:

- Extension to multivariate forecasts.
- Consider alternative online changepoint methods.